Advanced Quantum Mechanics

Leonard Susskind

Supplementary track, 2013 Fall
Transcript-derived notes curated by LazyingArt LLC

Original lectures by Leonard Susskind. Transcript-derived course notes curated by LazyingArt LLC.

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Chapter 1

Advanced Quantum Mechanics I: Quantum Dynamics and Symmetries

This chapter rebuilds the basic mathematical spine of the lecture: quantum states as vectors, particles in the position representation, the Hamiltonian-generated unitary time flow, and the symmetry principle that identifies conserved quantities by commuting operators. The goal is to move from abstract postulates to the concrete rules used in practical calculations, with a consistent convention and clear links between infinitesimal generators and finite transformations.

1.1 Review of the quantum kinematic structure

A physical system is represented by vectors in a (complex) vector space. A state is written as a ket $|\psi\rangle$, and its dual by $\langle\psi|$. The inner product $\langle\phi|\psi\rangle$ is a complex number associated to each pair of states.

$$\langle\psi|\psi\rangle = 1 \quad \text{for a normalized state.} \tag{1.1}$$

An observable A is represented by a Hermitian operator $\hat{A}$ ($\hat{A}^\dagger = \hat{A}$). The eigenvalue problem has physical meaning:

$$\hat{A}\,|a\rangle = a\,|a\rangle\,, \qquad a \in \mathbb{R}. \tag{1.2}$$

If the system is in $|a\rangle$, measurement of A is certain to give a. Operational distinguishability is encoded by orthogonality:

$$|\phi\rangle \text{ and } |\psi\rangle \text{ are distinguishable with certainty } \iff \langle\phi|\psi\rangle = 0. \tag{1.3}$$

In finite dimensions these are ordinary vectors; for particles and fields, the basis index becomes continuous and states are represented by functions.

1.2 Particle kinematics in the position representation

For a one-dimensional particle, define position eigenstates $|x\rangle$ by

$$\langle x|x'\rangle = \delta(x - x'). \tag{1.4}$$

The corresponding wavefunction is the position amplitude

$$\psi(x) = \langle x|\psi\rangle. \tag{1.5}$$

Its modulus squared is the position probability density:

$$p(x) = |\psi(x)|^2 = \psi^*(x)\psi(x), \tag{1.6}$$

and for normalizable states $\int dx\, |\psi(x)|^2 = 1$.

The position operator acts by multiplication:

$$(\hat{x}\psi)(x) = x\,\psi(x). \tag{1.7}$$

Distributional caveat. The "vectors" $|x\rangle$ are not ordinary square-integrable functions. Their coordinate overlap is the Dirac delta, and the position eigenfunctions are idealized delta spikes rather than physical packet states. Physical states are square-integrable approximants (e.g., narrow Gaussians).

1.3 Momentum in the same representation

The momentum observable is represented by a differential operator:

$$(\hat{p}\psi)(x) = -i\hbar\,\frac{d\psi(x)}{dx}. \tag{1.8}$$

(If $\hbar = 1$, drop it from formulas.)

Momentum eigenstates satisfy

$$\hat{p}\phi_p(x) = p\,\phi_p(x). \tag{1.9}$$

A convenient representative is

$$\phi_p(x) = \frac{1}{\sqrt{2\pi\hbar}}\,e^{ipx/\hbar}, \tag{1.10}$$

which has $|\phi_p(x)|^2 =$ constant: momentum certainty implies complete delocalization in position. This is the uncertainty tradeoff in precise mathematical form.

1.4 Time evolution as a unitary transformation

Time evolution is represented by a family $U(t)$ of operators mapping each state to a later state:

$$|\psi(t)\rangle = U(t)\,|\psi(0)\rangle\,, \qquad U(0) = \mathbf{1}. \tag{1.11}$$

A symmetry of information requires inner products to be preserved:

$$\langle\phi|\psi\rangle = \langle U(t)\phi|U(t)\psi\rangle \implies U^\dagger(t)U(t) = \mathbf{1}. \tag{1.12}$$

So $U(t)$ is unitary.

Assuming continuity, for small ϵ,

$$U(\epsilon) = \mathbf{1} - i\epsilon H + \mathcal{O}(\epsilon^2). \tag{1.13}$$

Unitarity to first order gives

$$H^\dagger = H, \tag{1.14}$$

so H is Hermitian and identified as the Hamiltonian.

Now use

$$|\psi(t+\epsilon)\rangle = U(\epsilon)\,|\psi(t)\rangle \tag{1.15}$$

and rearrange:

$$\frac{|\psi(t+\epsilon)\rangle - |\psi(t)\rangle}{\epsilon} = -iH\,|\psi(t)\rangle + \mathcal{O}(\epsilon). \tag{1.16}$$

Hence

$$i\,\frac{d}{dt}\,|\psi(t)\rangle = H\,|\psi(t)\rangle\,. \tag{1.17}$$

For wavefunctions this is

$$i\hbar\,\frac{\partial\psi(x,t)}{\partial t} = H\psi(x,t). \tag{1.18}$$

Finite time is obtained by exponentiation:

$$U(t) = e^{-iHt/\hbar}. \tag{1.19}$$

Stationary states are eigenstates of H:

$$H\,|E_n\rangle = E_n\,|E_n\rangle\,. \tag{1.20}$$

Then

$$|E_n(t)\rangle = e^{-iE_n t/\hbar}\,|E_n\rangle\,, \tag{1.21}$$

so the same Hamiltonian controls both time dependence and allowed energy levels.

1.5 Symmetry as commuting operators

A symmetry V is taken to be a unitary map on states that preserves all physical distinctions. If it represents a true symmetry of dynamics, then evolving then transforming is equivalent to transforming then evolving:

$$V\,U(t) = U(t)\,V, \qquad \forall t. \tag{1.22}$$

Using $U(t) = e^{-iHt/\hbar}$, this is equivalent to

$$[V, H] = 0. \tag{1.23}$$

For a continuous family $V(\epsilon) = \mathbf{1} - i\epsilon G + \mathcal{O}(\epsilon^2)$, the same logic gives

$$[G, H] = 0. \tag{1.24}$$

When $G = G^\dagger$, this is exactly a conserved generator:

$$\frac{d}{dt}\langle G\rangle = \frac{i}{\hbar}\langle [H, G]\rangle = 0. \tag{1.25}$$

Thus symmetry $\leftrightarrow$ conserved quantity is the operator-level statement.

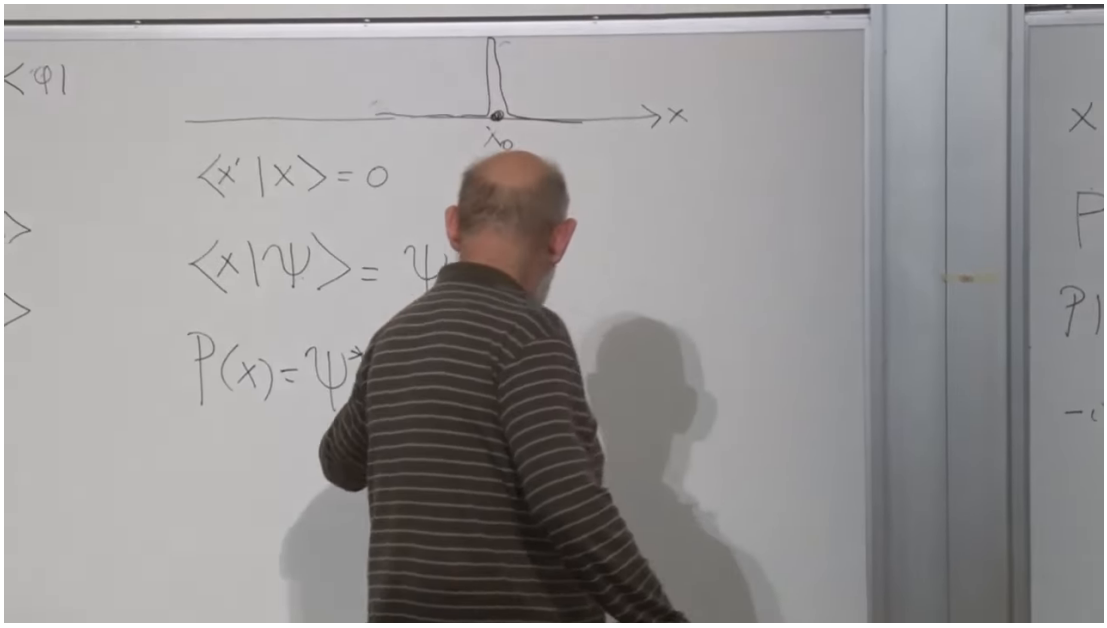


Figure 1.1: Infinitesimal-to-finite evolution structure for $U(\epsilon)$.

$$
\begin{tikzcd}
|\psi_1\rangle \arrow[r, "U(t)"] \arrow[d, "V"'] & |\psi_2\rangle = U(t)\,|\psi_1\rangle \arrow[d, "V"] \\
V\,|\psi_1\rangle \arrow[r, "U(t)"] & V\,|\psi_2\rangle = U(t)\,V\,|\psi_1\rangle
\end{tikzcd}
$$

Figure 1.2: A symmetry operation V commutes with time evolution $U(t)$ if the two routes between states coincide.

1.6 Continuous symmetry example: spatial translations

Define the spatial translation operator $T(\epsilon)$ acting on wavefunctions by

$$(T(\epsilon)\psi)(x) = \psi(x+\epsilon), \tag{1.26}$$

which is a rightward shift by ϵ. For small ϵ,

$$\psi(x+\epsilon) = \psi(x) + \epsilon\,\partial_x\psi(x) + \mathcal{O}(\epsilon^2), \tag{1.27}$$

so

$$T(\epsilon) = \mathbf{1} + \epsilon\,\partial_x + \mathcal{O}(\epsilon^2) = \mathbf{1} - \frac{i\epsilon}{\hbar}\,\hat{p}_x + \mathcal{O}(\epsilon^2). \tag{1.28}$$

Hence

$$T(\epsilon) = \exp\left(-\frac{i}{\hbar}\epsilon\,\hat{p}_x\right), \tag{1.29}$$

so the generator is

$$G_{\text{trans}} = \frac{\hat{p}_x}{\hbar}. \tag{1.30}$$

For a free particle,

$$H = \frac{\hat{p}_x^2}{2m}, \tag{1.31}$$

and $[H, \hat{p}_x] = 0$, so translation is a symmetry and $\hat{p}_x$ is conserved. Thus the same operator appears as both "generator of translations" and "conserved quantity," illustrating the lecture's central symmetry–conservation link.

1.7 Summary

The lecture's core chain is: state vectors → continuous basis amplitudes → operator representation → unitary time flow. Continuity plus unitarity forces $U(t)$ to be generated by a Hermitian Hamiltonian and yields the Schrödinger equations. A symmetry is an additional unitary map compatible with $U(t)$, which becomes a commutator condition with H. For continuous symmetries, infinitesimal generators G define finite transformations via exponentials, and commuting generators are conserved quantities. Translations provide the prototype: $T(\epsilon) = e^{-i\epsilon\hat{p}_x/\hbar}$, and for $H = p^2/2m$, momentum conservation follows directly from $[H, \hat{p}_x] = 0$.

Chapter 2

Symmetry, Degeneracy, and Angular Momentum for Quantum Rotational Systems

This chapter develops a symmetry-first view of rotational quantum systems. We define symmetries as operators commuting with the Hamiltonian, distinguish generic level coincidences from symmetry-protected multiplets, and build angular-momentum structure from the commutator algebra of generators. The particle on a circle provides an explicit worked model where quantized m-values, reflection pairing, and ladder operators emerge from operator algebra rather than ad hoc assumptions.

2.1 Symmetry as an invariance principle

A symmetry is an operation that leaves the quantum description unchanged. For a unitary transformation S, invariance of dynamics is the condition

$$SHS^\dagger = H$$

equivalently

$$[H, S] = 0.$$

Because S commutes with H, Schrödinger evolution is covariant under S: if $|\psi(t)\rangle$ evolves by H, then $S|\psi(t)\rangle$ evolves in the same way. In particular, E-eigenvalues are mapped to the same E-eigenspaces.

Symmetries can be exact (invariance of the full Hamiltonian), approximate (effective in a regime), or accidental (valid only for very specific parameter choices). Our focus is on exact symmetries because they control spectral structure.

2.2 Symmetry examples and the organizing role of operators

The same formal rule $[H, S] = 0$ unifies many familiar examples:

- spatial translations and rotations,

- interchange of identical particles,
- relabeling symmetries of a periodic lattice (discrete translation in crystal index),
- discrete mirror or inversion operations.

In practice, this is less about geometry alone and more about algebraic leverage: a set of commuting operators gives you a good quantum-number basis, while noncommuting families often generate nontrivial multiplets. Symmetry language gives a compact way to classify allowed degeneracies before solving differential equations.

Figure 2.1: Prototype symmetry motifs: geometric and discrete transformations that can commute with the Hamiltonian.

2.3 Degeneracy, symmetry, and when degeneracy is protected

For a bounded discrete spectrum, exact equality of two nearby energies is rare unless there is structure behind it. We call this *accidental* coincidence when no symmetry explanation exists. In contrast, symmetry-protected degeneracies are stable under small perturbations that preserve the same symmetry.

Proposition 2.1. *Let A and B be Hermitian symmetry generators commuting with H: $[A, H] = [B, H] = 0$. Define*

$$C := \frac{1}{i\hbar}[A, B].$$

Then C is Hermitian and $[C, H] = 0$.

Proof. Since A and B are Hermitian, $[A, B]^\dagger = -[A, B]$, so C is Hermitian. Then

$$[C, H] = \frac{1}{i\hbar}([AB, H] - [BA, H])$$

and expanding each commutator gives zero because both A and B commute with H. Hence $[C, H] = 0$. $\square$

Thus one can grow a closed set of symmetry generators by commutators. If that set closes to a Lie algebra with nontrivial structure constants, the representation theory is what drives degeneracy and spectral multiplets.

2.4 Particle on a circle: quantized L_z from single-valued wave functions

Let $\theta \in [0, 2\pi)$ label the position of a particle constrained to a circle. A small rotation by ϵ acts as

$$(\mathcal{R}(\epsilon)\psi)(\theta) = \psi(\theta - \epsilon).$$

Its infinitesimal form defines the generator L_z:

$$\mathcal{R}(\epsilon)\psi(\theta) \approx \left(1 - \frac{i\epsilon}{\hbar} L_z\right)\psi(\theta), \qquad L_z = -i\hbar\frac{d}{d\theta}.$$

Hence

$$L_z \left|\psi_m\right\rangle = m\hbar \left|\psi_m\right\rangle \quad \Longleftrightarrow \quad -i\hbar\frac{d\psi_m}{d\theta} = m\hbar\psi_m$$

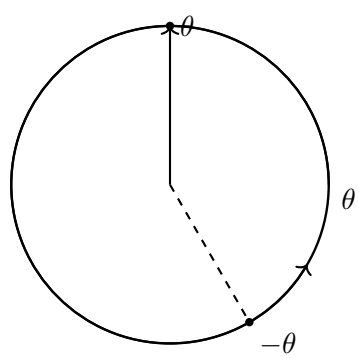
and therefore

$$\psi_m(\theta) = \frac{1}{\sqrt{2\pi}} e^{im\theta}.$$

Single-valuedness on $0 \sim 2\pi$ gives

$$\psi_m(\theta + 2\pi) = \psi_m(\theta) \iff e^{i2\pi m} = 1 \iff m \in \mathbb{Z}.$$

So the orbital angular-momentum quantum numbers are $m = 0, \pm 1, \pm 2, \ldots$.



Identification of θ and $\theta + 2\pi$: $\psi(\theta + 2\pi) = \psi(\theta)$

Figure 2.2: Circle coordinate, opposite angles, and periodicity condition.

2.5 Mirror symmetry and the $m \leftrightarrow -m$ pairing

Introduce a mirror (reflection) operator $\mathcal{M}$ with

$$(\mathcal{M}\psi)(\theta) = \psi(-\theta).$$

Its action on a momentum eigenmode is

$$\mathcal{M}\psi_m(\theta) = \psi_{-m}(\theta).$$

On a circle eigenstate:

$$\mathcal{M}L_z\psi_m = m\hbar\,\mathcal{M}\psi_m, \qquad L_z\mathcal{M}\psi_m = -m\hbar\,\mathcal{M}\psi_m,$$

so

$$[\mathcal{M}, L_z]\psi_m = 2m\hbar\,\mathcal{M}\psi_m \neq 0 \quad (m \neq 0).$$

Thus $\mathcal{M}$ does not commute with L_z, even though both may commute with H in a symmetric setup. When both symmetries are present (no mirror-breaking field), states m and $-m$ are paired at equal energy.

Important warning from the formalism: equal probabilities $|\psi_m(\theta)|^2 = |\psi_{-m}(\theta)|^2$ does not imply identical wave functions. Degeneracy concerns states and operators, not only densities.

A uniform magnetic field is a standard counterexample: it can split $\pm m$ levels while preserving some rotational structure of orbital motion in other settings. Symmetry arguments apply to what the Hamiltonian does, not to geometric intuition alone.

2.6 Generators, commutator algebra, and closure

For continuous symmetries a small transformation is written as

$$U(\epsilon) = 1 - \frac{i\epsilon}{\hbar}G + \mathcal{O}(\epsilon^2),$$

with Hermitian generator G. The commutator of two generators is another generator of another direction in the same symmetry structure if closure holds.

A useful worked pattern is:

$$[\,\mathcal{M}, L_z\,] \neq 0, \qquad \text{while } \; [H, L_z] = 0, \; [H, \mathcal{M}] = 0 \text{ (if mirror symmetric)}.$$

Hence $\mathcal{M}$ and L_z together generate two operations whose noncommutativity forces additional structure.

2.7 Three-dimensional angular momentum and ladder structure

For a particle in 3D,

$$L_x = yp_z - zp_y, \qquad L_y = zp_x - xp_z, \qquad L_z = xp_y - yp_x,$$

with canonical $[x_i, p_j] = i\hbar\delta_{ij}$. A compact expansion gives

$$[L_x, L_y] = i\hbar L_z,$$

and cyclic permutations. Equivalently,

$$[L_i, L_j] = i\hbar \sum_k \epsilon_{ijk} L_k.$$

For a rotationally invariant Hamiltonian, $[H, L_i] = 0$, so L_x, L_y, L_z are conserved quantities.

Define raising and lowering operators

$$L_\pm := L_x \pm iL_y.$$

Using the algebra above,

$$[L_z, L_\pm] = \pm\hbar L_\pm, \qquad [L_+, L_-] = 2\hbar L_z.$$

If $|\ell, m\rangle$ is an eigenvector of L_z,

$$L_z|\ell, m\rangle = m\hbar|\ell, m\rangle,$$

then

$$L_z(L_\pm|\ell, m\rangle) = (m \pm 1)\hbar(L_\pm|\ell, m\rangle),$$

so $L_\pm$ create a state with adjacent m-value when nonzero.

In a finite ladder representation, the chain terminates:

$$L_+|\ell, \ell\rangle = 0, \qquad L_-|\ell, -\ell\rangle = 0,$$

giving $m = -\ell, -\ell+1, \ldots, \ell$, hence $2\ell+1$ states in one rotational multiplet. For orbital wave functions on configuration space, $\ell, m \in \mathbb{Z}$ and $m \in \{-\ell, \ldots, \ell\}$. For more general angular-momentum types (including spin), half-integer m spectra are also possible; in this lecture the orbital circle argument selects the integer branch.

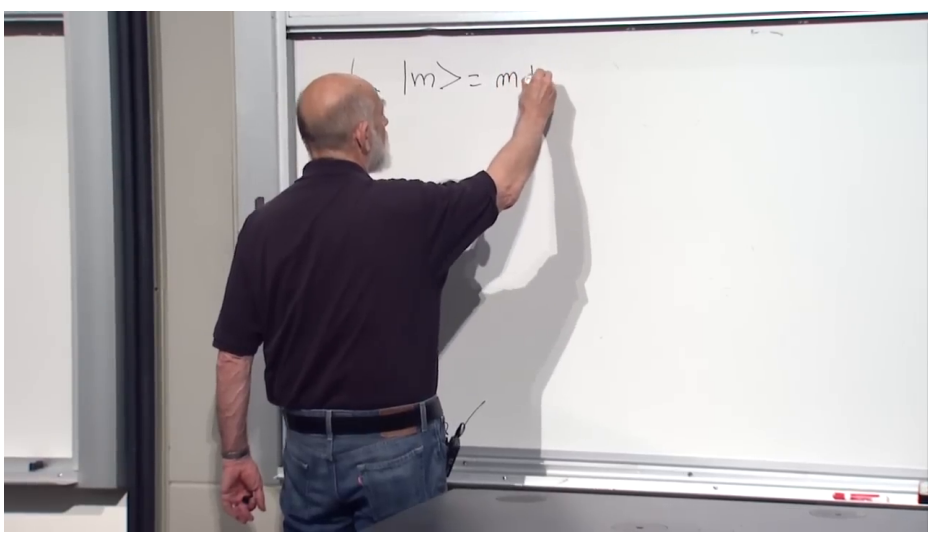


Figure 2.3: Illustrative ladder picture: $L_\pm$ connect adjacent m-levels inside a symmetry multiplet.

2.8 Summary

Symmetry enters as a commutation criterion with the Hamiltonian. Single symmetries can explain invariant spectral structure, but genuine multiplets require a noncommuting symmetry set whose commutator algebra closes. The circle provides the cleanest demonstration: rotational invariance gives $L_z = -i\hbar\partial_\theta$, single-valuedness gives integer m, and mirror symmetry links m to $-m$. In three dimensions, the algebra $[L_i, L_j] = i\hbar\epsilon_{ijk}L_k$ yields ladder operators $L_\pm$, a finite m-tower for each ℓ, and equal-energy multiplets in rotationally invariant systems.